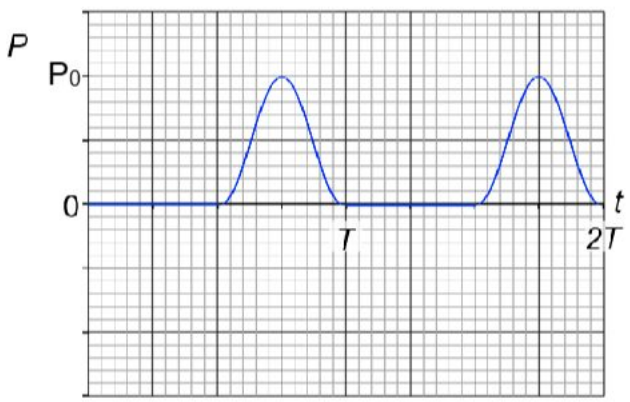
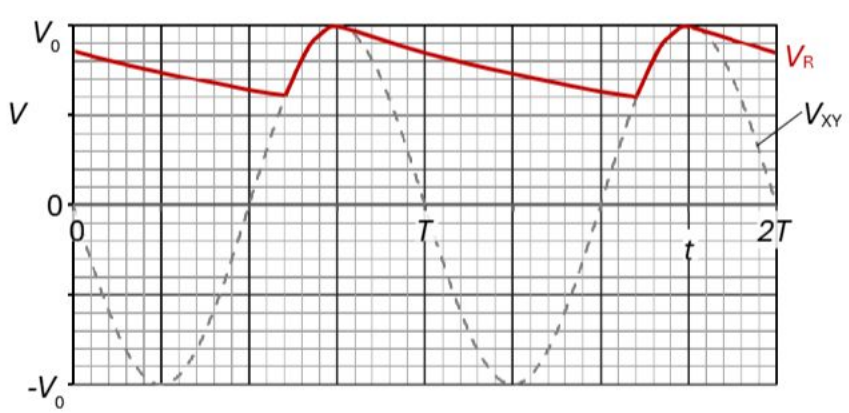
Question 6		
(a)(i)	Two sections of zero power of width $0.5T$ starting at $t = 0$ and $t = T$ Two humps of width $0.5T$ and peak P_0 *starting* at $t = 0.5T$ and $t = 1.5T$ [Deduct only this mark if the positions of the zero power and humps are swapped.] Correct sinusoidal-squared shape for the 2 non-zero humps (i.e. with smooth troughs seating on the t-axis at $P = 0$  	**B1** **B1**
(a)(ii)	The mean power within each hump is $\frac{1}{2}P_0$ from the symmetry of the curve. An additional half factor is from the removal of half of the power in each cycle (resulting in the ¼ factor).	**B1** **B1**
(a)(iii)	$\frac{V_{OUT,rms}^{\ 2}}{R} = \frac{1}{4}P_0 = \frac{1}{4}(2\frac{V_{XY,rms}^2}{R})$ $V_{OUT,rms} = \frac{V_{XY,rms}}{\sqrt{2}} = \frac{12}{\sqrt{2}} = 8.485 = 8.5$ V	**M1** **A1**
(b)(i)	Correct shape.  Note: No penalty if the first part of the graph is not drawn.	**B1**
(b)(ii)	From the graph, the time for the capacitor to discharge from V_0 to 0.60 V_0 is $0.85T$. $V = V_0 e^{-\frac{t}{\tau}} \Rightarrow \tau = \frac{-0.85T}{\ln 0.60} = 1.7T$	**B1** **A1**